



Secrets of Europa

Specialised probes or 'tunnelbots' will carry instruments below the icy layer of the Jupiter moon, Europa. <https://digit.in/jan19-67>



Water turned into fuel

Scientists designed an electrocatalytic material that acts like a mammalian lung to turn water into fuel. <https://digit.in/jan19-68>

D

inosaurs are charismatic megafauna that capture our imagination with their size and variety. The

dinosaurs were wiped out, along with 70 percent of all life forms, at the end of the Cretaceous period, about 66 million years ago. Pterosaurs, which were flying reptiles and the mosasaurs, carnivorous aquatic lizards also died out during these events. Both of these are equally charismatic megafauna, but are not dinosaurs, although they lived during the same time. One of the biggest unanswered questions in science is what exactly led to their extinction. Over the years, there have been many proposed explanations.

These theories include the newly evolved caterpillars wiping out forests because of a lack of natural predators,

THE METEOR STRIKE

The first person to suggest that an asteroid impact could have wiped out the Earth was MW De Laubenfels in 1956. The starting point for the theory was the mysterious Tunguska event of 1908. Considering the damage caused by the unknown object, De Laubenfels calculated the consequences of a much a larger strike. According to the paper, the dinosaurs were wiped out by "hot winds" of a large meteor strike.

In 1980, a team of researchers led by Nobel Laureate, Luis Walter Alvarez published a blockbuster paper. In the paper, the researchers announced the finding of a global layer of iridium in the geological record. Iridium is an abundant element in the bodies of the solar system, including asteroids. Most of the iridium in the Earth is expected to be deposited within its core, and

into the atmosphere as dust. This dust would spread across the world, and eventually settle down everywhere. The dust would then block out the sun, wiping out the bottom of the food chain - plants and phytoplankton in the oceans that produce oxygen. Only an asteroid about 10 kilometers in diameter striking the Earth would be able to cause such an impact. The impact would cause a crater about 100 kilometers in diameter.

Unknown to the researchers, such a crater had been discovered just two years before the paper was published. Geophysicists surveying the Gulf of Mexico for possible oil drilling locations noticed a vast crater embedded deep within the continental crust. The crater was 150 kilometers in diameter, expected to be caused by an object between 10 to 15 kilometers in diameter. It was reliably dated to within five million years of the extinction event that wiped out the dinosaurs, and 70 percent of all other species. The Chicxulub crater was the "smoking gun" for what is called the Alvarez hypothesis. The asteroid strike theory has since then grown to be the most widely accepted cause for the Cretaceous-Paleogene (K-Pg) extinction even, previously known as the Cretaceous-Tertiary (K-T) extinction event.

THE ERUPTIONS IN THE DECCAN FLATS

A link between volcanism and the K-Pg extinction event was first proposed by a 1978 paper by DM McLean. According to Dewey McLean, volcanic eruptions first put particulate matter into the atmosphere, which led to a period of global cooling. The organisms began adapting to the cooling, when a runaway greenhouse gas effect led to a sudden period of global warming. The organisms could not evolve rapidly enough to survive this sudden development, which is why most of the life forms on Earth died out.

Since then, there have been a small number of scientists holding on to the volcanism hypothesis. In a 2003 paper, Vincent Courtillot and Paul



Artist's impression of the impactor

allergic reactions to the emergence of flowering plants, a global pandemic, dinosaurs going blind because of exposure to UV radiation, and diseases spread by insects. These theories have been discredited because it cannot explain why some species survived while others died out, or why the marine reptiles died off too. This leaves two contenders for the extinction event.

very little is in the crust. The researchers concluded that the iridium layer, now called the iridium anomaly, was deposited by an extraterrestrial object - most likely an asteroid. The layer also has shocked quartz. Substructures in the crystals were knocked out of place by seismic waves travelling around the globe. About 60 percent of the mass of the object would be injected